

Spirillary Rat-Bite Fever (Sodoku) — Comprehensive Disease Characterization

Disease: Spirillary Rat-Bite Fever | **MONDO:** MONDO:0020532 | **Category:** Infectious Disease
Causative agent: *Spirillum minus* (syn. *Spirillum minor*) **Report date:** 2026-09-25

Evidence-source note. This is a bacterial zoonosis, not a genetic disorder. Sections designed for Mendelian/genetic diseases (host causal genes, pathogenic variants, inheritance, penetrance, chromosomal abnormalities, germline/somatic status) are **Not applicable**; where relevant, the analogous *pathogen* genetics/biology is provided instead. Evidence is predominantly **human clinical** (case reports/series, narrative and systematic reviews) with some **veterinary/microbiological** data. Much of the *Spirillum minus*-specific literature is old and observational because the organism cannot be cultured on artificial media; contemporary molecular data are sparse.

1. Disease Information

Overview. Spirillary rat-bite fever (RBF) is a systemic, relapsing febrile zoonosis acquired chiefly through the bite or scratch of a rat (or other rodent) infected with *Spirillum minus*, a small, tightly-coiled, motile, Gram-negative spiral bacterium. It is one of the two classic forms of rat-bite fever; the other, streptobacillary RBF, is caused by *Streptobacillus moniliformis*. The spirillary form predominates in Asia — its Japanese name "**sodoku**" (鼠疫, "rat poison") reflects this — whereas the streptobacillary form predominates in the Americas and Europe. RBF "is a rare but potentially fatal zoonotic disease caused by *Streptobacillus moniliformis* or *Spirillum minus*. In Asia, it is often caused by *Spirillum minus*" (P 41480582). It has been recognized as a clinical entity for over 2000 years (P 28002119).

Distinguishing clinical fingerprint (vs streptobacillary form). Spirillary RBF classically features a longer incubation (typically 1–4 weeks; often >7 days, vs 3–10 days for streptobacillary), **re-activation and ulceration of the healed original bite wound, regional lymphangitis/lymphadenopathy**, a **relapsing (recurrent) fever** pattern, and a distinctive violaceous/roseolar-to-macular rash; **arthritis and myalgia are uncommon** (arthritis is a hallmark of the streptobacillary form). It is **not** associated with the "Haverhill fever" foodborne/ingestion syndrome (that is streptobacillary).

Key identifiers. - **MONDO:** MONDO:0020532 (spirillary rat-bite fever) - **ICD-10:** A25.0 (Spirillosis — the spirillary form of rat-bite fever); A25.9 (rat-bite fever, unspecified); parent A25 (Rat-bite fevers) - **ICD-11:** 1B94 (Rat-bite fever) with the spirillary form as *Spirillum minus* infection - **MeSH:** "Rat-Bite Fever" (D011906); organism "*Spirillum*" (D013183) - **SNOMED CT:** Rat-bite fever (disorder); Spirillosis / *Spirillum minus* infection - **OMIM / Orphanet:** Not applicable as a Mendelian entry. Orphanet does not classify this common-cause infectious disease as a rare disease with an ORPHA code (RBF is generally handled through ICD, not Orphanet). - **NCBI Taxonomy (pathogen):** *Spirillum minus* — historically classified in genus *Spirillum*; note the type species *Spirillum volutans* is the validly-described genus member. *S. minus* has never been formally cultured/validly published under the Bacteriological Code, so its taxonomic status is provisional.

Synonyms / alternative names. Sodoku; spirillosis; spirillary fever; spirillum fever; *Spirillum minus* infection; *Spirillum minor* infection; rat-bite fever (spirillary type); (historically) "the Asian form" of rat-bite fever. Older literature also used *Spirochaeta morsus muris*.

Information source. Aggregated disease-level knowledge from case reports, case series and reviews; there is no large individual-patient (EHR) cohort resource for this rare disease.

HPO/ontology anchors for the entry: disease MONDO:0020532; see §3 for HP terms.

2. Etiology

Primary cause (infectious). Infection with *Spirillum minus*, a Gram-negative, aerobic, helically-coiled (2–5 tight spirals), rigid, highly motile bacterium (2–5 µm) bearing bipolar polytrichous flagella. Transmission is by **bite or scratch of an infected rat** (most common), other rodents (mice, squirrels), or animals that prey on rodents (cats, weasels, dogs). Unlike the streptobacillary form, **foodborne (ingestion) transmission is not described for *S. minus***. Human-to-human transmission does not occur. RBF is transmitted "to humans by" rodents and

their predators
([P 39628725](#)),
and can even be established "with contaminated vehicle contact alone, not only as a direct result of a bite"
([P 26584844](#)),
described for *S. moniliformis*).

Reservoir. Wild and domestic (pet/laboratory) rats are the principal reservoir; *S. minus* colonizes the oropharynx/nasopharynx and conjunctiva of a substantial fraction of healthy rats asymptotically. A comprehensive review covers the two causal species, their host species, pathogenicity (virulence factors and host susceptibility), diagnosis, therapy, epidemiology, transmission and prevention
([P 19008054](#)).

Haverhill fever distinction. The **foodborne** variant, *Haverhill fever*, "is a form of *S. moniliformis* infection believed to develop after ingestion of contaminated food or water"
([P 19008054](#))
— it is **exclusively streptobacillary** and has **no spirillary counterpart** (*S. minus* is not transmitted by ingestion).

Risk factors (environmental / behavioral). - **Rodent exposure** — the dominant risk factor. In a systematic review of streptobacillary endocarditis, "Exposure to rats was noted in 71.8% of patients, with 56.4% recalling a rat bite"
([P 37101553](#)).

- **Occupational:** laboratory-animal workers, pet-shop/feeder-rodent handlers, veterinarians, farm and sanitation workers, biomedical researchers. "Those working in places where rodents breed or are at risk of contact with rats or mice might be at risk"
([P 26584844](#)).

- **Pet ownership** — pet rats are an increasingly common source
([P 17223620](#));
([P 26701936](#));

"three domestic rats living in the girl's home"). "Keeping rats as pets cannot be recommended"

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P 34672901
).

- **Housing/poverty & crowding** — historically linked to higher rat contact ("housing conditions and habits of the people,"

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P 19867970

homelessness

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P 38459199
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- **Demographics:** children and low socioeconomic groups are a vulnerable population

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P 34672901
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- **Host immune status:** immunocompromise (cirrhosis, CKD, HIV/AIDS) predisposes to severe/complicated disease

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P 29709962
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P 11139161

RBF has presented as culture-negative septic arthritis in newly diagnosed HIV

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P 21877181
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Alcohol-use disorder is a recurrent comorbidity

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P 20397505
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Genetic host risk / protective factors. Not applicable / none established — no host germline susceptibility or protective variants, GWAS loci, or gene–environment interactions have been described for RBF. Susceptibility is essentially exposure-driven.

Protective factors (environmental). Rodent avoidance/control, wound hygiene and prompt post-bite wound care, use of gloves/protective equipment by animal handlers, and prompt post-exposure antibiotics (see §13).

CHEBI/agent anchor: causative agent *Spirillum minus* (NCBITaxon).

3. Phenotypes (Clinical Manifestations)

RBF "is characterized by a clinical triad of symptoms, fever, rash and arthritis" (

P 39267964

); however the **spirillary** form has a characteristic pattern that differs from the streptobacillary form. Frequency figures below are qualitative/derived from case literature (no large prospective cohort exists).

Phenotype	Type	HP term (suggested)	Onset/ course	Frequency in spirillary RBF
Fever, relapsing/ recurrent (spikes to 39–40°C, recurring every 3–5 days over weeks)	Symptom/ sign	HP:0001954 (Recurrent fever); HP:0001945 (Fever)	Subacute onset after 1–4 wk incubation; relapsing	Very frequent (hallmark)
Ulceration/ reactivation of the healed bite wound (indurated, painful, may eschar)	Clinical sign	HP:0200041 (Skin ulcer); HP:0025276 (Eschar)	Appears with first febrile relapse	Frequent (characteristic of sodoku)
Regional lymphangitis & lymphadenopathy proximal to bite	Clinical sign	HP:0002716 (Lymphadenopathy); HP:0100763 (Lymphangitis)	With fever onset	Frequent
Rash — violaceous/ red-brown macular, roseolar or maculopapular, classically petechial- purpuric on the extremities , and may involve palms and soles with mixed maculopapular/ pustular eruptions (P 21358889) P 9856201)	Physical manifestation	HP:0000988 (Skin rash); HP:0007398 (Maculopapular exanthema); HP:0000979 (Purpura); HP:0000967 (Petechiae); HP:0006740 (Palmoplantar rash, approximate)	With febrile relapses	Common (though may be absent — P 41480582)
Constitutional: malaise, headache, rigors/chills, myalgia	Symptom	HP:0002027; HP:0002315; HP:0025143; HP:0003326	With fever	Common

Phenotype	Type	HP term (suggested)	Onset/ course	Frequency in spirillary RBF
Arthralgia/arthritis	Symptom/ sign	HP:0002829; HP:0001369	Later	Uncommon in spirillary form (contrasts with streptobacillary, where migratory polyarthritis is typical)
Localized cellulitis (atypical presentation)	Sign	HP:0100658 (Cellulitis)	Days after bite	Reported (P 41480582)
Laboratory: neutrophilia, thrombocytopenia, elevated CRP	Lab abnormality	HP:0011897; HP:0001873; HP:0011227	Acute	Reported (P 41480582)
Laboratory: anemia, leukocytosis	Lab abnormality	HP:0001903; HP:0001974	Variable	Common in RBF (P 37101553): anemia 57%, leukocytosis 52%)
False-positive syphilis serology (VDRL/RPR reactive; treponemal tests negative)	Lab abnormality	HP:0031385 (abnormal serology, approximate)	During illness	Classic for spirillary RBF
Severe/complicated: endocarditis, meningitis, hepatitis, myocarditis, sepsis	Sign	HP:0100584; HP:0001287; HP:0200119; HP:0100806	Late/ untreated	Rare but high- mortality

Age of onset / severity / progression. Any age; children over-represented among reported cases. Severity ranges mild → life-threatening; untreated disease is **relapsing** and can persist for weeks to months, occasionally 1–2 years, with febrile bouts separated by afebrile intervals.

Quality-of-life impact. No formal EQ-5D/SF-36/PROMIS data exist for this rare disease. Acute illness causes days–weeks of incapacitating fever and malaise; treated disease usually resolves without sequelae ("recovered fully"/"without long-term sequelae,"

P 41480582

P 26701936

Complicated disease (endocarditis, osteomyelitis, septic arthritis) can cause lasting disability or death.

4. Genetic / Molecular Information

Host genetics — Not applicable. No causal genes, pathogenic variants, modifier genes, epigenetic signatures, or chromosomal abnormalities in the human host. Not heritable.

Pathogen molecular biology (analogous section). *Spirillum minus* is a fastidious spiral bacterium that **has never been grown in axenic culture**, so its genome is poorly characterized; there is no validated reference genome, and identification historically depended on morphology and animal passage rather than sequencing. This contrasts with *S. moniliformis*, for which 16S rRNA gene sequencing is the mainstay of molecular identification

P 34583654

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Consequently, no defined virulence genes, toxins, or resistance determinants are catalogued for *S. minus*; penicillin susceptibility is inferred clinically (§12).

5. Environmental Information

- **Infectious agent:** *Spirillum minus* (spiral Gram-negative bacterium). Reservoir: rats/rodents and their predators.
- **Environmental factors:** rodent-infested dwellings, poor sanitation, and settings with high human–rodent contact (urban poverty, homelessness, agriculture, laboratories, pet trade). Urbanization and climate change are increasing human–rodent interactions and zoonotic

risk generally

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P 41011829
).

- **Lifestyle factors:** keeping pet rodents; handling feeder rodents; sleeping in rodent-infested environments (bites often occur at night on hands/face of sleeping infants).
- **No toxic/radiation/chemical etiology.**

6. Mechanism / Pathophysiology

Causal chain (initiating event → clinical manifestation)

1. **Rodent bite/scratch (or contact with rodent secretions)** inoculates *Spirillum minus* into the skin and subcutaneous tissue → local infection. (*Demonstrated: transmission by bite.*)
2. The inoculation wound **initially heals**, then after an incubation of ~1–4 weeks the organism **re-activates locally**, producing induration, ulceration and sometimes eschar at the original bite site. (*Characteristic of sodoku; mechanism of the healing-then-reactivation is inferred, not molecularly demonstrated.*)
3. Organisms **drain via lymphatics** → **regional lymphangitis and lymphadenopathy**. (*Inferred from clinical pattern.*)
4. **Lymphohematogenous dissemination** → intermittent **bacteremia/spirochetemia**, which drives the **relapsing fever** (febrile bouts coinciding with waves of bacteremia; afebrile intervals as the host immune response transiently clears circulating organisms). (*Inferred, analogous to other relapsing bacteremic infections.*)
5. Circulating organisms and the host **innate inflammatory response** (neutrophilia, elevated CRP, cytokine release) → systemic constitutional symptoms and the characteristic **exanthem** (probably an immune/vasculitic-type cutaneous reaction). RBF has been reported to mimic and induce vasculitis, incl. ANCA/anti-endothelial antibodies

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P 37450033
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and to mimic Henoch–Schönlein purpura

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P 29709962
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implicating **immune-mediated small-vessel injury** in the skin.

6. In untreated/immunocompromised hosts, persistent bacteremia seeds distant sites → **metastatic/deep infection**: infective **endocarditis** (valve vegetations;

P 23993005

P 37101553

and (more typical of the streptobacillary form) septic arthritis, osteomyelitis, meningitis, hepatitis, myocarditis, abscesses → organ failure and, in ~7–13% untreated, **death**

P 17223620

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Branch point: After step 4, most immunocompetent, treated patients follow a benign self-limited/curable course (branch A); a minority — often with valvulopathy or immunocompromise (branch B) — progress to endocarditis and disseminated disease with high mortality

P 37101553

risk in prior valve disease

P 1562665

immunocompromise

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Mechanistic detail / ontology anchors

- **Cellular processes:** acute inflammation (GO:0006954, inflammatory response), innate immune response (GO:0045087), neutrophil-mediated response (GO:0002446), possible immune-complex/vasculitic tissue injury.
- **Cell types (CL):** neutrophils (CL:0000775), macrophages (CL:0000235), vascular endothelial cells (CL:0000115), lymphocytes.
- **Immune involvement:** predominantly innate + humoral; agglutinating antibodies to the organism arise during infection (historically documented,

P 19867970

"production of powerful agglutinins for the organism"). Reactive autoantibodies (ANCA, AECA) can appear

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P 37450033

- **Tissue damage mechanisms:** direct bacterial invasion + immune-mediated small-vessel/endothelial injury; valve vegetation formation in endocarditis.
- **Molecular pathways:** classical innate pattern-recognition/inflammatory signaling (e.g., TLR–NF-κB, IL-1/IL-6/TNF cytokine axes) — **inferred by analogy**, not specifically demonstrated for *S. minus*.
- **Molecular/omics profiling, single-cell, CRISPR screens, etc.: Not available** — the organism is unculturable and rare; no transcriptomic/proteomic/metabolomic datasets exist. Even for the culturable *S. moniliformis*, "Virtually nothing is known regarding prevalence in humans and animal reservoirs," and full-genome studies of *Streptobacillus* are only now being called for to define virulence traits

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P 27088660

— for *S. minus* the genomic knowledge gap is essentially total.

7. Anatomical Structures Affected

- **Primary site:** skin/subcutaneous tissue at the bite (UBERON:0002097 skin of body; UBERON:0002199 integumentary system) and **regional lymphatic vessels/nodes** (UBERON:0001473 lymph node; UBERON:0001473; lymphatic vessel UBERON:0001473→UBERON:0005802).
- **Body systems involved:** integumentary, lymphatic/hematologic (bacteremia), and — in complications — **cardiovascular** (endocardium/heart valves, UBERON:0002165 endocardium; UBERON:0000946 cardiac valve), **musculoskeletal** (joints UBERON:0000955→UBERON:0002544; bone), **nervous** (meninges, UBERON:0002360), **hepatic** (liver UBERON:0002107), and reticuloendothelial (spleen — splenomegaly reported,

P 37101553

- **Tissue/cell level:** vascular endothelium and dermal small vessels (rash/vasculitis); valvular endocardial surface (vegetations); synovium and bone in metastatic infection.

- **Subcellular:** no specific organelle target (extracellular bacterium); GO cellular component anchors are not specifically implicated.
 - **Localization / laterality:** bite-site lesion and lymphadenopathy are **regional/unilateral** (side of the bite); rash and systemic features are generalized/bilateral. Endocarditis most often affects the **mitral valve**, then aortic > tricuspid > pulmonary
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P 37101553
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8. Temporal Development

- **Onset:** acquired at any age; **subacute** onset after an **incubation of ~1–4 weeks** (spirillary form characteristically longer than the 3–10 days of streptobacillary RBF; presumptive diagnosis relies on "incubation period,"
- (
P 41480582
).
- **Course:** untreated disease is **relapsing/episodic** — recurrent febrile bouts every few days with afebrile intervals — persisting for weeks to months, occasionally 1–2 years.
 - **Progression rate:** variable; usually self-limited-to-chronic if untreated in the immunocompetent, but can be rapidly progressive/fatal if endocarditis or sepsis supervenes.
 - **Remission:** prompt penicillin produces rapid, usually complete resolution (treatment-induced remission); spontaneous resolution can occur but relapse is common without treatment.
 - **Critical window:** early antibiotic therapy (before endocarditis/dissemination) is the key intervention period — "Early initiation of empiric antibiotic therapy ... can prevent serious complications"
- (
P 41480582
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9. Inheritance and Population (Epidemiology)

- **Inheritance:** Not applicable (infectious; no heritability, penetrance, expressivity, anticipation, mosaicism, founder effect, consanguinity, or carrier frequency).

- **Incidence/prevalence:** true incidence unknown; RBF is **rare, under-reported and under-diagnosed**

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P 29671179);

P 39628725).

Rat bites are frequently **not a mandatory-notification event**, so no registry exists — in Buenos Aires "there is no record of it," and 50% of bitten slum residents sought no care

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P 40793882).

The largest single Canadian series found **11 cases on Vancouver Island over 2010–2016**

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P 31015812).

The spirillary form is rarer than the streptobacillary form outside Asia.

- **Geographic distribution:** spirillary RBF (**sodoku**) is concentrated in **Asia** — Japan, China, India and elsewhere — "In Asia, it is often caused by *Spirillum minus*"

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P 41480582);

first presumptive *S. minus* case in Nepal reported 2026

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P 41480582);

a Kenyan case with spirillum-like organisms on thick film

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P 1286642).

The streptobacillary form predominates in the Americas/Europe. The greater historical frequency in Japan was attributed to housing conditions and rat exposure

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P 19867970).

- **Demographics:** children and people of low socioeconomic status are over-represented

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P 34672901);

occupational cohorts (lab/pet-rodent handlers) at risk. Endocarditis series skew male (61.5%) with mean age ~41

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P 37101553).

- **Sex ratio / age:** no strong intrinsic sex predisposition for uncomplicated disease (exposure-driven); bimodal exposure in young children (bites) and adult handlers.

10. Diagnostics

The central diagnostic challenge: *S. minus* cannot be cultured on artificial media, and RBF's symptoms are nonspecific, so diagnosis is frequently **clinical/presumptive**

([P 41480582](#))

"the importance of clinical suspicion over microbiological confirmation").

- **Direct microscopy (mainstay for *S. minus*):** demonstration of the characteristic tight spiral, motile organism by **darkfield microscopy** or **Giemsa/Wright stain** of blood, bite-wound exudate, or aspirate of the regional lymph node. A Kenyan case was diagnosed by "demonstration of spirillum like organisms from a thick blood film"

([P 1286642](#))

- **Animal inoculation:** intraperitoneal inoculation of blood/exudate into mice or guinea pigs, then examining the animal's blood/peritoneal fluid for spirilla — a classic method when microscopy is negative.
- **Molecular: 16S rRNA gene PCR/sequencing** from blood, joint or tissue is the most sensitive modern test and can identify RBF organisms in culture-negative cases

([P 34672901](#))

used for *Streptobacillus* speciation in

([P 34583654](#))

([P 40472936](#))

([P 35365242](#))

Note a caveat: 16S rRNA analysis "may be uncertain for proper pathogen identification" and RBF diagnostics remain a "diagnostic dilemma"

([P 27088660](#))

Metagenomic next-generation sequencing (mNGS) can detect the pathogen directly from clinical samples "in less than 72 h" even when "blood culture results are negative"

([P 31315559](#))

MALDI-TOF MS is used for *Streptobacillus* but **not** reliable for the unculturable *S. minus*.

- **Blood culture:** typically **negative** for *S. minus*

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P 41480582);

positive cultures instead indicate the streptobacillary form (BACTEC/subculture;

P 29709962
P 23993005).

Even for *S. moniliformis*, routine cultures often fail "because of the fastidious nature of the organism's growth, as well as inhibitors present in standard blood culture bottles" — specifically **sodium polyanethol sulfonate (SPS)** anticoagulant; the organism may grow only in SPS-free media such as thioglycolate broth

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P 20397505).

Prolonged incubation and notifying the lab of suspected RBF are advised.

- **Serology:** no standardized serologic assay; **false-positive non-treponemal syphilis tests (VDRL/RPR)** occur in a substantial minority of spirillary cases (treponemal tests negative) — a useful diagnostic clue.

- **Laboratory:** neutrophilia, thrombocytopenia, elevated CRP

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P 41480582);

anemia (~57%), leukocytosis (~52%), raised inflammatory markers (~58%) across RBF

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P 37101553).

- **Imaging:** echocardiography (TEE) for suspected endocarditis

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P 23993005);

MRI for osteomyelitis/discitis

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P 34039283
P 40472936).

- **Genetic testing (WGS/WES/panels/karyotype/CMA/repeat testing):** Not applicable (no host genetic disease).

- **Omics-based diagnostics (of the pathogen):** unbiased **metagenomic** NGS of blood, pus or tissue is an emerging, rapid, culture-independent confirmatory test

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P 31315559);

host transcriptomic/ proteomic/metabolomic diagnostic signatures are **not established** for RBF.

Diagnostic criteria: no formal society criteria exist; diagnosis rests on **compatible clinical syndrome + rodent-exposure history + supportive microscopy/PCR** (or response to penicillin).

Differential diagnosis: streptobacillary RBF, leptospirosis, borreliosis/relapsing fever, secondary syphilis (owing to false-positive RPR), Rocky Mountain spotted fever and other rickettsioses, meningococemia, infective endocarditis of other cause, disseminated gonococcal infection, viral exanthems, malaria, reactive/rheumatoid arthritis, ANCA-associated vasculitis

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P 37450033

and

Henoch–Schönlein

purpura

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11. Outcome / Prognosis

- **Untreated mortality ~7–13%** — concordant across sources: "mortality rate of 7% to 10% if untreated"

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P 21358889

"up to 13%, if untreated"

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P 34211343

and "10%"

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- **With prompt appropriate antibiotics: excellent** — full recovery, usually without sequelae

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"full clinical recovery when treated in a timely and appropriate manner").

- **Complications (predominantly untreated/immunocompromised):** infective endocarditis (rare but frequently fatal — **death in 36%** of pooled cases, surgery in 36%,

P 37101553

sepsis, metastatic septic arthritis/osteomyelitis/discitis

P 34039283

P 40472936

P 26948832

meningitis

P 34583654

hepatitis, myocarditis, abscess formation, and a **hyperinflammatory syndrome resembling hemophagocytic lymphohistiocytosis (HLH)** with multiorgan failure

P 32868746

- **Prognostic factors:** speed of diagnosis/treatment, presence of endocarditis or pre-existing valve disease

P 1562665

immune status (cirrhosis/CKD/HIV worsen outcome —

P 29709962

P 11139161

and age/comorbidity.

- **QoL/disability measures:** no formal instruments reported; residual disability mainly from musculoskeletal or cardiac complications.

12. Treatment

Suggested NCIT anchors given where applicable.

- **First-line pharmacotherapy: Penicillin** (NCIT:C716) — IV penicillin G for severe/systemic disease, or oral penicillin/amoxicillin for mild disease; typically 7–14 days (uncomplicated), longer for deep infection. *S. minus* is penicillin-susceptible; "The bacterium is generally susceptible to penicillin antibiotics with full clinical recovery"

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P 26701936
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Patients are often "cured completely after intravenous administration of penicillin G" with therapy "completed by an oral course of doxycycline"

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P 9856201
).

Case cured with penicillin ($\pm$ gentamicin) —

P 1286642 .

- **Alternatives (penicillin allergy): doxycycline** (NCIT:C641)/tetracyclines, **ceftriaxone** (NCIT:C1737)/cephalosporins, or macrolides. Combination ampicillin + doxycycline used in the Nepal *S. minus* case

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P 41480582
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ampicillin/sulbactam + doxycycline

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P 38459199
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- **Endocarditis/deep infection:** high-dose IV penicillin (often with an aminoglycoside such as **gentamicin**, NCIT:C557) for 4–6 weeks; ceftriaxone-based regimens also used

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P 23993005
)

— IV penicillin 6 wk + gentamicin 2 wk;

P 11139161

— ceftriaxone/gentamicin/penicillin).

- **Aminoglycoside adjunct:** gentamicin as synergistic add-on in severe disease

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P 1286642
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P 23993005

- **Surgical/interventional:** wound debridement for local disease

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P 41480582
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valve replacement/surgery in ~36% of endocarditis (10/14 needing surgery had replacement,

P 37101553);

arthroscopic lavage/DAIR for septic arthritis / prosthetic joint infection

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P 40472936
P 35365242).

- **Supportive care:** antipyretics/analgesia, fluids, treatment of sepsis; NSAIDs for arthralgia/vasculitic features

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P 37450033).

- **Caution:** a **Jarisch–Herxheimer reaction** may follow the first antibiotic doses (as with other spiral-organism infections) — monitor and support.
- **Pharmacogenomics, gene/cell/RNA/targeted/immunotherapy: Not applicable.**
- **Treatment response:** high cure rates with timely therapy; poorer with delayed diagnosis or endocarditis.

13. Prevention

- **Primary prevention: rodent control** and avoidance; safe handling of pet/laboratory rodents (gloves, avoid face/hand contact, secure housing); public/occupational **health education**; discourage keeping rats as pets, especially around young children ("keeping rats as pets cannot be recommended,"

P 34672901).

- **Post-exposure wound care & prophylaxis:** immediate cleansing/irrigation of bite wounds; consideration of **post-bite antibiotic prophylaxis** (e.g., a penicillin/amoxicillin course) after high-risk rodent bites; tetanus prophylaxis as indicated.
- **Secondary prevention:** early clinical suspicion in febrile patients with rodent exposure and prompt empiric antibiotics to prevent endocarditis/dissemination

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P 41480582).

- **Tertiary prevention:** complete antibiotic course + follow-up echocardiography where endocarditis is a concern; source control (debridement, valve surgery) as needed.
- **Immunization: none available** (no human vaccine).
- **Public health / One Health:** integrated rodent surveillance and control; the burden of rodent-borne zoonoses is rising with urbanization and climate change, warranting

interdisciplinary One Health approaches

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P 41011829
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- **Genetic counseling / screening / newborn screening: Not applicable.**

14. Other Species / Natural Disease

- **Pathogen taxonomy:** *Spirillum minus* (NCBITaxon; genus *Spirillum*).
- **Reservoir hosts (NCBI Taxon):** *Rattus norvegicus* (NCBITaxon:10116, Norway/brown rat), *Rattus rattus* (NCBITaxon:10117, black/house rat), *Mus musculus* (NCBITaxon:10090) and other murids; carnivores that prey on rodents (cats, dogs, weasels) can transmit via bite.
- **Natural disease in animals:** rats usually carry the organism **asymptotically** (oropharyngeal/ conjunctival colonization). Clinically apparent rodent disease is better documented for related *Streptobacillus* species — e.g., *Streptobacillus notomys* caused fatal otitis interna/media with neurologic signs in house rats

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P 29671179
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and evidence suggests host tropism (*S. notomys* with *R. rattus* vs *S. moniliformis* with *R. norvegicus*). Analogous systematic carriage of RBF agents in pet/feeder murids is globally documented

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P 37643287
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- **Zoonotic potential / cross-species susceptibility: high zoonotic potential** — humans are infected from the rodent reservoir; there is no human-to-human spread. Rodents are recognized reservoirs for RBF among many bacterial zoonoses

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P 41011829
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- **Comparative biology / orthologous host genes: Not applicable** (no host disease gene).

15. Model Organisms

- **Classic in vivo model / diagnostic bioassay: mouse** (*Mus musculus*, MGI) and **guinea pig** (*Cavia porcellus*) intraperitoneal **inoculation** was historically used both to propagate *S.*

minus (which cannot be grown on media) and as a **diagnostic** tool (examine animal blood/peritoneal fluid for spirilla). This is an **induced/experimental infection** model rather than a genetic model.

- **Natural animal host as model: rats** (*Rattus* spp., RGD) are natural carriers and have been used to study colonization and transmission; *S. notomys* rat infection provides a natural-disease model of *Streptobacillus* pathogenicity

([P 29671179](#)).

- **Genetic models (knockout/transgenic/etc.): Not applicable** — no host genetic disease to model; the pathogen is unculturable, precluding standard genetic/CRISPR manipulation and axenic experimental systems.
- **Model limitations:** because *S. minus* cannot be cultured, controlled reproducible molecular studies are essentially impossible; most mechanistic understanding is inferred from clinical observation and by analogy to *S. moniliformis* and other spiral bacteria. Koch's postulates were historically only partially fulfilled for RBF organisms

([P 19867970](#)).

- **Resources:** MGI, RGD (host animals); no dedicated *S. minus* strain repository exists because the organism is not maintained in culture.

Summary Answer

Spirillary rat-bite fever (**sodoku**, MONDO:0020532) is a rare, under-diagnosed, **relapsing febrile zoonosis caused by *Spirillum minus***, a non-culturable spiral Gram-negative bacterium transmitted mainly by the bite/scratch of infected rats (predominant in Asia). It classically presents after a 1–4-week incubation with reactivation/ulceration of the healed bite wound, regional lymphadenopathy, a relapsing fever, and a violaceous rash (arthritis is uncommon, distinguishing it from the streptobacillary form); diagnosis is largely clinical/presumptive with microscopy, animal inoculation or 16S rRNA PCR because blood cultures are negative. It is not a genetic disease—there are no host causal genes, inheritance, or heritable risk factors—and it is **highly treatable with penicillin** (untreated mortality ~7–13%, rising sharply with complications such as endocarditis), with prevention resting on rodent control, safe rodent handling, and prompt post-bite wound care and antibiotics.

Limitations and Future Directions

Evidence limitations. - **No high-level evidence.** The entire knowledge base is case reports, small case series, and narrative/ systematic reviews of those reports — there are **no RCTs, cohort studies, or registries** for RBF, so incidence/prevalence, true frequency of each phenotype, and comparative treatment efficacy are all imprecise. - **Spirillary-specific data are especially thin.** Because *S. minus* **cannot be cultured**, most modern molecular literature concerns *S. moniliformis*; several claims here (incubation length, relapsing-fever mechanism, false-positive RPR, arthritis rarity) rest on classic/older observational descriptions and textbook consensus rather than contemporary primary data, and some cited quotes describe the streptobacillary form and are extended to the spirillary form by analogy (flagged in-text). - **Ontology/identifier caveats.** ICD-11 code (1B94) and several UBERON/HP mappings are best-available suggestions; the ICD-10 A25.0 "Spirillosis" mapping is the firmest spirillary-specific anchor. *S. minus* has no validly published bacteriological name or reference genome. - **Mechanism is largely inferred.** The pathophysiology causal chain is reconstructed from clinical phenomenology plus analogy to other spirochetal/relapsing infections; molecular pathways, cytokine profiles, and cell-type contributions have **not** been directly demonstrated for *S. minus*.

Future directions. - Full-genome sequencing of *Streptobacillus* strains and any culturable/enrichable *S. minus* material to define virulence traits and improve identification (P 27088660).

- Systematic prevalence/carriage surveys in rodent reservoirs and exposed human populations.
- Broader deployment of **16S rRNA PCR and metagenomic NGS** to capture culture-negative and spirillary cases and to build a molecularly-confirmed case series (P 31315559).
- Making rodent bites/RBF a notifiable event to enable real epidemiologic estimates (P 27088660).
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Supported vs refuted (framing for this descriptive task). - **Supported:** infectious (non-genetic) etiology; penicillin curability; ~7–13% untreated mortality; culture-negativity/diagnostic difficulty; rodent-exposure and immunocompromise as risk factors; endocarditis as

the key lethal complication. - **Refuted/negative:** no host genetic causal or susceptibility loci; no vaccine; no foodborne (Haverhill) route for the spirillary form; no established omics/ biomarker diagnostic signature.

Key References (PMIDs)

41480582 (S. minus RBF, Nepal, 2026) · 37450033 (RBF/vasculitis; S. minus morphology) · 39267964 (RBF triad) · 39628725 (RBF genus review) · 17223620 (RBF review; 10% untreated mortality) · 26701936 (penicillin cure) · 37101553 (systematic review, streptobacillary endocarditis; 36% mortality) · 34672901 (case-report review; 16S PCR; pet-rat caution) · 1286642 (spirillum thick-film diagnosis, Kenya) · 26584844 (non-bite transmission) · 29671179 (S. notomys natural rat disease) · 41011829 (rodent zoonoses / One Health) · 19867970 (historical etiology) · 28002119 (RBF known >2000 yrs) · 23993005, 11139161, 1562665 (endocarditis) · 29709962 (immunocompromised/HSP-mimic) · 34583654, 40472936, 35365242, 34039283, 38459199, 26948832, 32868746 (complications & molecular diagnosis) · 19008054 (comprehensive RBF review; Haverhill fever) · 21358889 (pediatric RBF; mortality 7–10%; acral rash) · 34211343 (rodent carriage; mortality up to 13%) · 34813430 (relapsing fever/rash) · 40793882 (non-notifiable; vulnerable-population survey) · 31015812 (Vancouver Island case series) · 27088660 (RBF/Streptobacillus diagnostics review; 16S limitations; prevalence unknown) · 31315559 (metagenomic NGS diagnosis) · 9856201 (pet-rat RBF; penicillin G + doxycycline cure; palmoplantar rash) · 20397505 (culture pitfalls/SPS inhibitor; thioglycolate broth) · 21877181 (RBF in HIV/AIDS; culture-negative septic arthritis).

Evidence base is dominated by case reports/series and reviews; Spirillum-minus-specific molecular data are scarce because the organism cannot be cultured.